

Quotient Laws of Exponents

Teacher's Edition

The Period

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational-number bases.

Must Whole-number exponents with rational bases; the zero exponent is derived from the quotient law, not announced.

Must not Bases are nonzero wherever a quotient or a zero exponent appears, and 0^0 is never asked. Negative exponents wait for 3.04.

Target I can use the quotient of powers, power of a quotient and zero exponent laws to rewrite and evaluate a numerical expression.

MTR **MTR.5.1** — Notes III · **MTR.6.1** — Board 3 · **MTR.4.1** — Board 5

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-6	11 min
Example 1	7-8	3 min
Your Turn	9-10	3 min
Example 2	11-12	3 min
Your Turn	13-14	3 min
Whiteboards	15-32	13 min
Independent Set	33	6 min
IXL	34	5 min
Total		53 min

Say these three things out loud today

1. Same base, subtract the exponents — top minus bottom, in that order.
2. An exponent on a fraction lands on the top and on the bottom, both.
3. Anything nonzero to the zero power is 1. Not $0 - 1$.

Slide by Slide

1. **Quotient Laws of Exponents.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the move the whole lesson runs on: cancel a factor of 5 from top and bottom, and one 5 is left.
3. **Warm-Up — Answers.** Reveal. 1. 3^5 2. -3 3. $x < 4$ 4. 5
4. **Notes — Where the rule comes from.** Subtract, do not divide. 5^6 over 5^3 is 5^3 , not 5^2 . Question 6 in the round has 4^4 waiting for the divider.
5. **Notes.** Three copies of $8/10$, multiplied: $8 \cdot 8 \cdot 8$ over $10 \cdot 10 \cdot 10$.
6. **Notes.** Zero exponent means NO factors of the base are multiplied in. What is left is the 1 you always start from.
7. **Example 1 — A negative base.** SPOKEN ONLY.
8. **Example 1 — Worked.** Subtract, keep the base with its sign, evaluate: three negative factors, negative answer.
9. **Your Turn — Same steps, your numbers. Boards up when done.** Catches a divider ($3^3 = 27$ from $6 \div 2$) and a student who wrote 1^4 by dividing the bases.
10. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
11. **Example 2 — A decimal base inside a quotient.** SPOKEN ONLY.
12. **Example 2 — Worked both ways.** Route 1 keeps the numbers small until the last step.
13. **Your Turn — Same steps, your numbers. Boards up when done.** Catches a student who cubes only the numerator ($27/4$) or only the denominator ($3/64$).
14. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
- 15-16. **Board 1.** $\frac{10^5}{10^3} \rightarrow 10^2$ 10^8 — added the exponents [confused with product of powers]; 1^2 — divided the bases [7.NSO.1.1 misconception: operating on the bases]
- 17-18. **Board 2.** A website puts 8^4 photos into 8 equal albums. The site is 30 days old. How many photos are in each album? $\rightarrow 512$ The 30 days is not needed. 4096 — never divided by the 8 albums [the quotient was not taken]; 64 — subtracted 4 — 1 wrong or used 8^2 ; anything built from 30 — the unneeded figure was used
- 19-20. **Board 3.** $\frac{(-6)^6}{(-6)^5} \rightarrow -6$ 6 — dropped the sign of the base; 1 — divided the bases [7.NSO.1.1 misconception]
- 21-22. **Board 4.** $\left(\frac{-5}{10}\right)^7 \rightarrow \frac{(-5)^7}{10^7}$ $(-5)^7/10$ — exponent applied to the numerator only [7.NSO.1.1 misconception: exponent applied to part of a base]; $(-0.5)^7$ — simplified first (correct value, but not what was asked)
- 23-24. **Board 5.** 9^0 Which is the value? $\rightarrow A - 1$ (B) confused the zero exponent with the identity exponent, 9^1 [7.NSO.1.1 law names]; (C) treated the zero exponent as a reciprocal, which is what a NEGATIVE exponent does [misconception carried into 8.NSO.1.3]; (D) zero exponent read as 'nothing', so 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

- 25-26. Board 6.** $\frac{4^8}{4^2}$ Which expression is equivalent? → **C — 4^6** (A) divided the bases as well as subtracting the exponents [7.NSO.1.1 misconception: operating on the bases]; (B) added the exponents — the product law applied to a quotient [confused with product of powers]; (D) divided the exponents, $8 \div 2$ [7.NSO.1.1 misconception: 'divide' applied to the exponents] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.
- 27-28. Board 7.** $\frac{0.5^8}{0.5^5}$ → **0.125** 0.5^{13} — added the exponents instead of subtracting [quotient of powers confused with product of powers]; 0.5^{40} — multiplied the exponents [confused with power of a power]
- 29-30. Board 8.** $\left(\frac{10}{11}\right)^0$ → **1** $10/11$ — identity exponent instead of zero exponent; 0 — zero exponent read as 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$]
- 31-32. Board 9.** Hannah cancels $2^3/2^3$ to 1. Sebastian subtracts to 2^0 . Who is right? → **Both — $2^0 = 1$** 'Hannah, because $2^0 = 0$ ' — zero exponent read as 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$]; 'Sebastian only' — did not accept that a nonzero number over itself is 1
- 33. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.
- 34. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

- Board 1.** 10^8 — added the exponents [confused with product of powers]
- Board 2.** 4096 — never divided by the 8 albums [the quotient was not taken]
- Board 3.** 6 — dropped the sign of the base
- Board 4.** $(-5)^7/10$ — exponent applied to the numerator only [7.NSO.1.1 misconception: exponent applied to part of a base]
- Board 5.** confused the zero exponent with the identity exponent, 9^1 [7.NSO.1.1 law names]
- Board 6.** divided the bases as well as subtracting the exponents [7.NSO.1.1 misconception: operating on the bases]
- Board 7.** 0.5^{13} — added the exponents instead of subtracting [quotient of powers confused with product of powers]
- Board 8.** $10/11$ — identity exponent instead of zero exponent
- Board 9.** 'Hannah, because $2^0 = 0$ ' — zero exponent read as 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$]
- Materials** Whiteboards and markers. Calculators are allowed on everything; $5^6 = 15,625$ is on the notes for the value check.